RAAGHUL NATARAJ KANNAN

Tempe, AZ 85280 | +16234981850 | raaghuln03@gmail.com | LinkedIn | GitHub

Skills

- **Languages** : Python, SQL, R
- **ML / AI**: NLP, PyTorch, TensorFlow, Scikit-learn, Feature Engineering, Model Evaluation, Hyperparameter Tuning, Reinforcement Learning, MLOps, Large Language Models, RAG, AI Agents
- **Data Engineering**: Apache Kafka, ETL Pipelines, Real-time Data Streaming, Neo4j, MongoDB
- **Cloud & DevOps**: AWS, Azure, Docker, Kubernetes, CI/CD Pipelines, REST APIs
- **Tools**: Git, Power BI, SHAP, Pandas, React, FastAPI

Experience

Sorim Technologies | ***AI Engineer Intern*** **Jan 2026 - Present**

- Built a production-grade AI-powered pharmaceutical operations platform enabling manufacturing engineers to query live plant data and operational documents using natural language, accelerating production insight discovery.
- Engineered a Retrieval-Augmented Generation (RAG) pipeline using LLMs, embeddings, and vector search to deliver grounded responses from live plant data and uploaded documents, reducing hallucinations.
- Designed and developed the full-stack application using React, FastAPI, and MongoDB, building secure REST APIs, document ingestion workflows, and real-time analytics dashboards.
- Developed domain-specific AI personas, voice-enabled interactions, and multi-turn conversational workflows, improving accessibility and usability for manufacturing teams.

Tech Mahindra | ***AI Engineer Intern*** **Jun 2025 - Aug 2025**

- Built and deployed a production NLP chatbot using Python, achieving 89% F1-score on intent classification and reducing customer support tickets by 32%.
- Designed low-latency REST APIs integrating AI inference services, improving average response latency by 25% under peak load
- Applied A/B testing framework to evaluate model variants across 10,000+ user sessions, informing data-driven decisions on model promotion

Ford Motor Company | ***Data Engineer Intern*** **Jan 2024 - May 2024**

- Engineered high-throughput ETL pipelines processing 1M+ rows of operational data daily, ensuring data integrity across downstream analytics systems
- Refactored SQL queries and applied indexing strategies in Power BI reports, cutting dashboard latency by 40% and accelerating executive decision cycles
- Developed Python data validation scripts with automated quality checks, raising reporting accuracy by 28% and eliminating ~3 hours of daily manual review

Flex | ***Machine Learning Engineer Intern*** **Feb 2023 - Jun 2023**

- Developed predictive maintenance models using scikit-learn with SHAP-based interpretability on Flex's cloud infrastructure; improved fault prediction accuracy by 15%, reducing unplanned downtime
- Conducted ANOVA-based statistical analysis on manufacturing process data, surfacing key process variation drivers and guiding process adjustments to reduce defect rates

Projects

Vision-Language Model for Satellite-Based Disaster Assessment **Jan 2026 - May 2026**

Arizona State University

- Led a 5-member team to develop a vision-language model for automated disaster damage classification using the xView2 and SpaceNet datasets.
- Developed a vision-language pipeline using PyTorch and Transformers to generate structured disaster assessment reports.
- Fine-tuned multimodal models for disaster damage classification and performance evaluation.

Real-Time Streaming Data Pipeline https://github.com/raaghul07/real-time-streaming-pipeline **Jan 2025 - Apr 2025**

Arizona State University

- Built containerized real-time streaming pipeline using Apache Kafka and Kubernetes (Helm) processing NYC taxi trip data at high throughput
- Implemented graph analytics on Neo4j including BFS pathfinding and PageRank to identify and rank high-traffic Bronx corridors
- Deployed full distributed stack with Zookeeper, Kafka-Neo4j connector, and Neo4j GDS using Kubernetes Helm charts; demonstrates end-to-end distributed systems design

Education

Arizona State University | ***Master of Science, Data Science*** **Aug 2024 - May 2026**

SRM Institute of Science and Technology | ***Bachelor of Technology, Artificial Intelligence*** **Sep 2020 - Jun 2024**

Certifications

AWS Certified Cloud Practitioner: **Oct 2026**

https://cp.certmetrics.com/amazon/en/public/verify/credential/b201832fc8134a72a2e991672dcb78d6

- Demonstrated understanding of AWS core services, including EC2, S3, Lambda, RDS, VPC, and IAM.
- Gained foundational knowledge of cloud security, shared responsibility models, high availability, scalability, and cost optimization.
- Developed familiarity with AWS analytics, AI/ML services, serverless computing, and cloud architecture principles.